FINAL AI/ML PROJECT REPORT

Customer Churn Prediction Using Machine Learning

6-Week Artificial Intelligence Internship

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Model: Random Forest Classifier

A complete machine learning workflow covering problem definition, data preparation, exploratory analysis, model development, prediction and evaluation.

1. Introduction

Customer churn is a major business challenge for subscription-based organizations. When customers discontinue a service, the organization loses recurring revenue and may also incur additional costs to acquire replacement customers. Machine learning can help identify patterns associated with churn and support proactive retention strategies.

This project develops a customer churn prediction system using the IBM Telco Customer Churn dataset. The workflow covers data cleaning, exploratory data analysis, feature encoding, model training and evaluation using a Random Forest Classifier.

2. Problem Statement

The objective is to predict whether a telecom customer is likely to churn based on demographic information, service usage, contract details and billing-related attributes. The target variable is **Churn**, with “Yes” representing churn and “No” representing customers who remain with the company.

3. Project Objectives

• Load and understand a real-world customer churn dataset.

• Clean missing and inconsistent values and remove unnecessary identifiers.

• Perform exploratory data analysis to identify patterns related to churn.

• Transform categorical variables into numerical features suitable for machine learning.

• Train a Random Forest classification model.

• Generate predictions and evaluate them using accuracy, precision, recall, F1-score and a confusion matrix.

• Interpret the results from a business perspective and identify future improvements.

4. Why AI/ML?

Customer churn is influenced by multiple factors that can interact in non-linear ways. Traditional rule-based approaches may not capture these relationships effectively. Machine learning can learn patterns from historical customer records and apply those patterns to unseen customers. The resulting predictions can be used as decision support for retention teams.

5. Dataset Description

The project uses the IBM Telco Customer Churn dataset. It contains customer demographic, service, contract and billing information, with Churn as the target variable. Important features include tenure, Contract, InternetService, MonthlyCharges, TotalCharges, PaymentMethod, OnlineSecurity and TechSupport.

The dataset contains approximately 7,000 customer records and 21 original columns. The customerID field is an identifier rather than a predictive feature and was removed before modeling.

6. Data Preprocessing

The TotalCharges column required numeric conversion because some values were stored as text/blank entries. These values were converted using numeric coercion and missing values were filled using the median. Duplicate records were removed. The customerID identifier was dropped.

Categorical predictor variables were converted to numerical columns using one-hot encoding. The target was mapped from No/Yes to 0/1. The data was then divided into training and testing sets using an 80:20 split with stratification to preserve the class distribution.

7. Exploratory Data Analysis

Five visualizations were created to understand the dataset and churn patterns: churn distribution, churn by contract type, churn by tenure, monthly charges by churn, and a numerical correlation heatmap.

Key observations included a noticeable class imbalance between customers who stayed and customers who churned. Month-to-month contract customers showed substantially more churn than customers on longer contracts. Customers with shorter tenure were also more likely to churn. Monthly charges showed differences between churn groups, while tenure and total charges displayed a positive relationship.

8. Model Development

A Random Forest Classifier was selected because it can capture non-linear relationships, handle interactions among multiple customer attributes and provide feature-importance estimates. The model was configured with 200 trees and a fixed random state for reproducibility.

The encoded feature matrix was used as X and the binary churn label was used as y. The model was fitted on the training data and predictions were generated for the unseen test data.

9. Feature Importance

Random Forest feature importance was calculated to identify which encoded attributes contributed most strongly to the model's decisions. This provides an interpretable view of the factors the model relied on and can help guide future feature engineering and retention analysis.

10. Model Evaluation

The model was evaluated on **1,409 unseen test customers**. Accuracy, precision, recall and F1-score were calculated for the churn classification task. A confusion matrix was also generated to examine correct and incorrect predictions by class.

Metric	Churn / Overall Result
Accuracy	79%
Precision (Churn)	64%
Recall (Churn)	50%
F1-Score (Churn)	56%
No Churn Precision	83%
No Churn Recall	90%
No Churn F1-Score	86%
Incorrect Predictions	294 / 1,409

11. Business Interpretation

The model achieved 79% overall accuracy, showing that it learned useful patterns from the customer data. However, recall for the Churn class was 50%. This means the model identified approximately half of the customers who actually churned. Precision of 64% means that when the model predicted churn, the prediction was correct in approximately 64% of cases.

For a retention application, false negatives are important because they represent customers who churned but were not identified as high risk. Therefore, improving churn recall should be a major priority in future iterations. The model can be used to prioritize customers for retention campaigns, personalized offers, loyalty programs and proactive support, while keeping human review in the decision process.

12. Error Analysis

There were **294 incorrect predictions** in the test set. These errors include false positives, where a customer was predicted to churn but did not, and false negatives, where a customer actually churned but was predicted not to churn. The relatively low 50% churn recall indicates that the model misses a meaningful portion of actual churners.

13. Limitations

• The model was evaluated on one train/test split rather than extensive cross-validation.

• The baseline Random Forest configuration was not extensively hyperparameter-tuned.

• Class imbalance can make overall accuracy look stronger than performance on the churn class.

• The model predicts based on historical attributes and does not directly observe future customer behavior.

• Feature importance does not by itself establish causal relationships.

14. Future Scope

• Use class weights or threshold tuning to improve churn recall.

• Perform cross-validation and systematic hyperparameter optimization.

• Compare Random Forest with Logistic Regression, Gradient Boosting, XGBoost or other ensemble models.

• Engineer behavioral features such as tenure bands, service combinations and billing patterns.

• Use probability-based risk scores instead of only Yes/No predictions.

• Deploy the model through a Streamlit dashboard or cloud application for interactive retention analysis.

15. Conclusion

This project successfully implemented an end-to-end customer churn prediction workflow. The IBM Telco dataset was cleaned and transformed, exploratory analysis was performed, and a Random Forest Classifier was trained to predict churn. The model achieved 79% accuracy, with 64% precision, 50% recall and 56% F1-score for the Churn class.

The results demonstrate that machine learning can provide useful customer-risk predictions, while also showing that further improvement is required to detect more actual churners. The project provides a practical foundation for developing a stronger, deployable customer retention system.

16. Technologies and Libraries

Python, Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn were used for data handling, visualization and machine learning. The project was developed in a Jupyter Notebook environment.

17. Final Workflow

Problem Definition → Data Collection → Data Cleaning → Data Analysis → Feature Engineering → Train/Test Split → Model Training → Prediction → Evaluation → Business Interpretation → Future Scope

References: IBM Telco Customer Churn dataset; Pandas documentation; NumPy documentation; Matplotlib documentation; Seaborn documentation; Scikit-learn documentation.